

# OCR-Rescue Architecture: Box vs. Crop Extraction Benchmark

VisionPsy-Nano-460M (stock) & VisionPsy-Nano-460M-Flash — On-Device via llama.cpp

Prepared for: VLM Tester project (Android, on-device ID document extraction)

Test date: 2026-09-18 | Device: Samsung Galaxy M35 5G (SM-M356B, Mali-G68 GPU, 8-core CPU)

Test data: 6 real Aadhaar cards | 48 total inference runs (24 per model) + 5-trial grounding probe + 6-trial prompt ablation

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## 1. Executive Summary

**33%**

Nano — box-case accuracy

**100%**

Nano — crop-case accuracy

**42%**

Flash — box-case accuracy

**92%**

Flash — crop-case accuracy

**The single most important finding of this report:** across two models and 48 total test runs on six real Aadhaar cards, physically **cropping** the image to the OCR-detected field and sending just that crop to the VLM is decisively more reliable than sending the **full image with a box drawn on it**. Crop-based extraction was correct on 23 of 24 runs (96%) combined across both models. Box-based extraction was correct on only 9 of 24 runs (37.5%) combined, and its failures follow one dominant, repeatable pattern: the model ignores the box entirely and defaults to extracting whatever field it was most heavily trained to extract (typically name, DOB, gender, or the Aadhaar number), regardless of what the box actually points to.

This finding directly overturns the original architecture assumption. The plan to have the OCR layer draw a bounding box on the full image and hand it to the AI layer with an instruction like *"look at the boxed area and extract data out of it"* is not a sound design for this model family. The OCR layer should instead crop the image to the detected field region (with a small context margin) and send only that crop.

Three supporting investigations back this conclusion and are documented in full below:

- **The model has no reliable grounding capability** (Part 2) — it cannot reliably map a text-described location or phrase to real coordinates, so any design that leans on the model interpreting coordinates is unsound regardless of the box/crop question.
- **The box-ignoring failure is not fixable by prompt engineering** (Part 9) — an explicit, forceful prompt that directly forbade the exact failure mode observed still failed on 5 of 6 attempts, no better than a plain instruction.
- **The failure is not a measurement artifact** (Part 4) — five of six box placements were found to have real pixel-level measurement issues (clipping digits, overlapping a QR code, sitting at the exact edge of the text) during this investigation. All were corrected and every affected case was re-tested. Box-case accuracy did not meaningfully improve after the fixes, confirming this is genuine model behavior, not a byproduct of imprecise boxes.

On latency: Flash is **not** meaningfully faster than the stock Nano model on this hardware (Part 8) — both average in the 55-70 second range per image depending on case, with the well-established root cause (vision-encoder tiling cost, not model size or reload time) applying equally to both.

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## 2. Background & Why This Investigation Happened

The project's original architecture change (requested mid-investigation) proposed: upon an OCR-layer extraction failure on a specific ID document field, the OCR layer would draw a bounding box on the full document image, compute that box's pixel coordinates, and hand both the boxed full image and the coordinates to the on-device VLM (VisionPsy-Nano-460M or its "Flash" variant) with a prompt along the lines of *"From this image, look at the boxed area and extract the data that was missed."* This was proposed specifically as a latency-reduction measure, since sending a reduced/targeted region was assumed to let the model focus its attention rather than parse the entire document.

Before implementing this in the app, three questions needed answering with evidence rather than assumption:

1. Does the model actually understand spatial/coordinate references at all (grounding capability)?

2. Does drawing a visible box on the image reliably restrict the model's reading to that region?
3. Is a box-annotated full image, or a physically cropped region, the more reliable and faster input?

This report documents the investigation and answers to all three, plus the antecedent latency root-cause work that motivated the whole redesign in the first place.

### 3. Part 1 — Latency Root-Cause Investigation (Recap)

Before any box/crop work began, the original complaint was that the 460M VisionPsy-Nano model was taking 100+ seconds per extraction on-device, comparable to or slower than a much larger 2B-class model (Gemma via LiteRT-LM) documented elsewhere in this project. The initial hypothesis — reinforced by externally-sourced advice pasted into the investigation — was that the dominant cost was **full model reload on every single inference run**, and that a persistent server/cache architecture would fix it.

**Finding: model reload is not the bottleneck.** A custom ARM build of `llama-server` was compiled from source (the project ships no host C++ toolchain, requiring a stand-in stub compiler for one build-tool dependency) and used to isolate load time from generation time. Measured model + mmproj load time was consistently **~1.7 seconds**. This is not the bottleneck.

**Finding: vision-encoding dominates the cost.** Server logs consistently showed the SigLIP2 vision encoder processing a fixed **13 image chunks / 887 total prompt tokens** for the test document, taking **63-89 seconds** depending on thread count — the overwhelming majority of total latency. Text generation itself took only 1.5-9 seconds. Example measured breakdown from a captured server log:

```
prompt processing (vision encode + prefill): 63163.46 ms / 887 tokens (14.04 tok/s)
generation:                               1523.80 ms / 42 tokens (27.56 tok/s)
total:                                     64687.26 ms / 929 tokens
```

**Finding: resizing the image does not reduce tile count.** The vision encoder uses an "AnyRes"/LLaVA-UHD-style resolution-candidate-grid selection mechanism that picks a tiling layout by **aspect ratio**, not raw pixel count. The same test image produced identical `n_chunks=13`, 887 total tokens at both 636×400 and 318×200 (same aspect ratio, 4x fewer pixels). Plain resizing is not a usable lever.

#### Dead ends closed with direct evidence (not assumption):

- `--image-max-tokens` / `--image-min-tokens` CLI flags exist and are correctly wired through to the model's parameter struct, but have **zero effect** on this model — the code path they gate is only active for a different projector type (hparams gate condition `image_min_pixels > 0` is never true for this "custom" SigLIP2 projector).
- GPU (Vulkan) offload was built and tested; it is **slower** than CPU on this device's Mali-G68 GPU, which lacks cooperative-matrix support.
- Thread routing was verified correct in source — the vision encoder receives the same thread count as the language model, it is not starved.
- The server's prompt/KV cache is a single unified cache keyed on image hash + token count (not separate per sub-model as originally suspected) — a cache hit skips both vision-encoding and LM prefill together, but only on an exact byte-identical image match.

Net effect of a persistent-server / avoid-reload architecture: saves the ~1.7s load time only. This is a real but small optimization (~2% of a typical 80s run), not the fix the original latency complaint needed. The only genuine levers found were thread count (raising from 2 → 4 cores measured 82s vs. 100+s in live app testing) and reducing input image size in a way that changes tile count — which is exactly what motivated testing actual crops rather than full images.

## 4. Part 2 — Does the Model Have Grounding Capability?

### 4.1 Definition used

**Grounding** (visual grounding / referring expression comprehension): the ability to map a natural-language description to a specific spatial region in an image — given a phrase such as "the Aadhaar number," the model outputs the bounding box coordinates of that region. This is the reverse of object detection, and it is a *specifically trained* capability requiring (image, region, phrase) triples during training — it does not emerge automatically from general captioning or field-extraction training.

This matters for the architecture because if the model cannot reliably map coordinates to visual content, any prompt design that relies on the model interpreting coordinates (in either direction) is unsound regardless of the box-vs-crop question.

### 4.2 Test methodology

Using the project's existing public sample ID image (non-PII test data), the on-device Nano model was asked, with no visible box or mark of any kind, to return the bounding box of a named field in the exact format `[x1,y1,x2,y2]` , having first been told the image's exact pixel dimensions (636 × 400).

### 4.3 Results

| Prompt                                                                                                                                               | Expected region      | Model output                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------|
| "...Return the bounding box of the Aadhaar number...in the exact format [x1,y1,x2,y2]..."                                                            | [263, 258, 468, 295] | <b>[374, 162, 968, 542] – x2 = 968 exceeds the image's actual 636px width. Physically impossible.</b>          |
| Same format, asking for "the person's name"                                                                                                          | [265, 148, 415, 178] | <b>"Of India" – abandoned the coordinate task entirely, returned an unrelated text fragment from the card.</b> |
| Same Aadhaar-number ask, using the exact <code>bbox_2d</code> - style JSON format the model had spontaneously produced in an earlier, unrelated test | [263, 258, 468, 295] | <b>"Of India" – same abandonment as above.</b>                                                                 |

### 4.4 Interpretation

Two of three attempts did not even produce a coordinate-shaped answer — the model defaulted to reading arbitrary text off the card instead, meaning it does not reliably recognize "produce a bounding box" as the requested task. The one attempt that did produce coordinates produced values that don't fit inside the image it was just told the dimensions of. This is consistent with the language model surface-pattern- matching on bounding-box-shaped text it

may have seen incidentally during pretraining (the observed value range is suggestive of a 0-1000 normalized coordinate convention used by some other grounding-trained models, e.g. Qwen2-VL), rather than deriving a real location from the vision encoder's actual output.

**Why this doesn't contradict the box-attention success seen elsewhere in this report:** There are two different capabilities being tested. *Grounding* requires interpreting an abstract symbol (a phrase or a number) with no direct pixel presence and deriving a location from it — nothing in this model's training supplies that skill. *Reading a drawn box* requires no such abstraction — the red rectangle is literal new pixel content that the vision encoder directly perceives, the same as any other real visual feature on the card. Basic visual salience and "read what's highlighted" instruction- following do not require grounding training. This model has the second capability (with caveats explored in Parts 5-8) but not the first.

**Conclusion: no usable grounding capability.** Any design that asks the model to interpret textual coordinates (either to find a region or to validate one) is not supported by this model. The OCR layer must do 100% of the localization; the model must only ever be shown, not told, where to look.

## 5. Part 3 — Box vs. Crop Benchmark: Methodology

### 5.1 Test data

Six real Aadhaar cards were used, each assigned one field to test (chosen to cover a spread of field types — a numeric ID, a date, a single English word, a name, and a full multi-line address):

| Card holder              | Field under test | Ground truth value                                                                                                              | Card format                                                                                  |
|--------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Meghna Kumari            | aadhaar_number   | 6606 2751 1012                                                                                                                  | Standard compact card (front+back)                                                           |
| Rameez Zaheer Kadiwala   | date_of_birth    | 12/07/2002                                                                                                                      | Full A4 enrollment letter (860×1912px — much taller/different aspect ratio than the other 5) |
| Prakash Gopichand Rathod | gender           | Male                                                                                                                            | Compact card, front+back stacked in one image                                                |
| Jasjot Singh             | name             | Jasjot Singh                                                                                                                    | Compact card, front only                                                                     |
| Aditiya Sahu             | aadhaar_number   | 3597 8269 8331                                                                                                                  | Compact card, front only                                                                     |
| Sapna Bajaj              | address          | W/O: Ajit Kumar Bajaj, Ward No 01, Seujpur, Gabhrupathar, OPP Tirupati Balaji Mandir, Dibrugarh West, Dibrugarh, Assam - 786001 | Standard compact card (front+back)                                                           |

Note: Aditiya's card was originally assigned `address`, but his uploaded image is front-only and has no address field printed anywhere on it. The field was reassigned to a second `aadhaar_number` sample (giving that field 3 independent samples), and `address` was moved to Sapna's card, which has a legible back-panel address block.

### 5.2 Test image generation — four variants per card

For each card, four test images were generated from the one ground-truth field box:

- **box\_clean** — the full original image with a red rectangle drawn exactly around the field.
- **box\_distorted** — the full image rotated -5°, sheared 0.07, and Gaussian-blurred (radius 1.3), simulating a crooked/lower-quality phone photo. The box's four corners were transformed through the identical affine matrix used to distort the image, so the drawn box continues to track the field's new position exactly.
- **crop\_clean** — the image cropped tightly to the field box plus a 14px context margin, no box drawn (the crop itself isolates the field).
- **crop\_distorted** — the clean crop rotated/sheared/blurred the same way (radius 1.0, slightly less blur given the much smaller image).



This produced 24 test images per benchmark run (6 cards × 4 variants), run identically against both models — 48 total inference calls for the main benchmark.

### 5.3 Prompts

Two fixed prompts were used for all box/crop queries respectively (temperature 0, max\_tokens 80, identical across every image and both models):

**Box case:** "There is a red rectangle/box drawn on this image. Output only the exact text that appears inside that red box. Do not output anything else - no labels, no explanation, no extra text."

**Crop case:** "Extract and output only the text visible in this image. Do not output anything else - no labels, no explanation, no extra text."

Note the crop prompt intentionally asks for "all text visible" rather than one named field, since the crop already isolates the field — any secondary text caught by the 14px context margin (e.g. an adjacent "VID:" line) is correctly included per this prompt's own instruction, and was not scored as an error.

### 5.4 Server configuration

Both models were served via a custom-built ARM `llama-server` binary running directly on the test device (Samsung Galaxy M35, SM-M356B), queried over HTTP via an `adb forward` USB tunnel. All inference computation (vision encoding and generation) executed on the phone's own CPU — the host machine only prepared images and sent/received HTTP requests.

```
./llama-server -m <model>.gguf --mmproj <mmproj>.gguf --port 8090 -t 4 -tb 4 --no-warmup
```

Nano: `visionpsy-nano-460m-q5_k_m.gguf + mmproj-visionpsy-nano-460m-q8.gguf`

Flash: `visionpsy-nano-460m-flash-q5_k_m.gguf + mmproj-visionpsy-nano-460m-flash-q8.gguf`

### 5.5 Scoring methodology

A response was scored **correct** if the ground-truth value's core content was present and accurate, tolerant of: digit-group spacing differences (e.g. "359782698331" vs "3597 8269 8331"), extra unrequested-but-present text in crop responses (per §5.3), and a garbled/incorrect Devanagari transliteration accompanying a correct English value (a known, separately-documented model quirk unrelated to this investigation). A response was scored **incorrect** if the field's actual value was wrong, missing, replaced by a different field's value, or the model produced unrelated/hallucinated content.

## 6. Part 4 — Box Measurement Corrections (Methodological Transparency)

During the benchmark, box placements for five of the six cards were found — through direct pixel-level inspection with a fine coordinate grid overlaid on each image — to have real measurement defects. Each was corrected and every affected test case was re-run on both models. This section documents exactly what was wrong, to be transparent about data quality and because the corrections themselves produced useful evidence (see callouts below).

| Card    | Defect found                                                                                    | Original box                          | Corrected box        | Effect on re-test                                                                                                                             |
|---------|-------------------------------------------------------------------------------------------------|---------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Aditiya | Box was drawn directly through the middle of the digits, visually obscuring parts of the number | (98, 195, 270, 219)                   | (115, 202, 289, 230) | <b>box_clean flipped from a 2-digit misread to an exact match on Nano</b>                                                                     |
| Jasjot  | Box sat at the exact boundary of the text on both left and right edges, no margin               | (270, 155, 420, 183)                  | (262, 151, 429, 187) | <b>box_distorted flipped from a garbled hallucination to a correct read on Nano</b>                                                           |
| Meghna  | Right edge clipped the final digit; bottom edge touched the "VID:" line below                   | (48, 210, 282, 240)                   | (140, 208, 305, 231) | box_clean failure persisted unchanged (name/DOB/gender leaked) — confirms this specific failure is model behavior, not the measurement defect |
| Sapna   | Right edge extended ~100px into the adjacent QR code pattern                                    | (435, 90, 800, 150)                   | (432, 88, 696, 148)  | box case failures persisted on both variants — confirms model behavior, not measurement                                                       |
| Prakash | Bottom-left corner touched a Devanagari diacritic mark, minor tightness                         | (215, 230, 335, 262)                  | (212, 229, 348, 271) | box case failures persisted unchanged on both models                                                                                          |
| Rameez  | None found — box was already clean with proper margin                                           | (285, 1547, 580, 1578)<br>– unchanged |                      | —                                                                                                                                             |

**Why this matters for the overall conclusion:** two of five corrections (Aditiya, Jasjot) did change individual results, proving the corrections were real and worth making. But three of five (Meghna, Sapna, Prakash) showed *no change* in the box-case failure after fixing the measurement — the model kept ignoring the (now perfectly placed) box in exactly the same way. This is the strongest evidence in this report that the box-case failure is a genuine, repeatable model behavior (a strong trained bias toward its default field-extraction pattern) rather than an artifact of sloppy box placement. All figures in Parts 5-8 below use the fully corrected boxes.

## 7. Part 5 — Full Results: Nano (Stock)

| Card    | Field          | Variant        | Latency | Model output                                                                                                                                                  | Verdict                                                          |
|---------|----------------|----------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Meghna  | aadhaar_number | box_clean      | 64.0s   | "Meghna Kumar shades, Mehna kumari, 23/10/1997, male/Female"                                                                                                  | ✗ box ignored                                                    |
| Meghna  | aadhaar_number | box_distorted  | 63.3s   | "6606 2751 1012"                                                                                                                                              | ✓                                                                |
| Meghna  | aadhaar_number | crop_clean     | 62.8s   | "6606 2751 1012"                                                                                                                                              | ✓                                                                |
| Meghna  | aadhaar_number | crop_distorted | 62.7s   | "6606 2751 1012"                                                                                                                                              | ✓                                                                |
| Rameez  | date_of_birth  | box_clean      | 53.3s   | "Rameez Zaheer Kadiwala"                                                                                                                                      | ✗ box ignored, gave name                                         |
| Rameez  | date_of_birth  | box_distorted  | 54.9s   | "612487298241"                                                                                                                                                | ✗ box ignored, gave Aadhaar number                               |
| Rameez  | date_of_birth  | crop_clean     | 32.1s   | "HANE LEANY OF TARA / DOB: 12/07/2002"                                                                                                                        | ✓ (DOB value correct; garbled Devanagari transliteration prefix) |
| Rameez  | date_of_birth  | crop_distorted | 56.3s   | "जनम तारिख / DOB: 12/07/2002"                                                                                                                                 | ✓                                                                |
| Prakash | gender         | box_clean      | 80.7s   | "Prakash Gopichand Rathod"                                                                                                                                    | ✗ box ignored, gave name                                         |
| Prakash | gender         | box_distorted  | 82.2s   | "The answer is: 8538 7535 4900"                                                                                                                               | ✗ box ignored, gave Aadhaar number                               |
| Prakash | gender         | crop_clean     | 60.5s   | "पुरsey / Male"                                                                                                                                               | ✓ (garbled Devanagari prefix, English value correct)             |
| Prakash | gender         | crop_distorted | 87.0s   | "hells / Male"                                                                                                                                                | ✓                                                                |
| Jasjot  | name           | box_clean      | 75.8s   | "Jasjot Singh"                                                                                                                                                | ✓                                                                |
| Jasjot  | name           | box_distorted  | 91.3s   | "Jasjot Singh"                                                                                                                                                | ✓                                                                |
| Jasjot  | name           | crop_clean     | 62.6s   | "Jasjot Singh"                                                                                                                                                | ✓                                                                |
| Jasjot  | name           | crop_distorted | 62.6s   | "Jasjot Singh"                                                                                                                                                | ✓                                                                |
| Aditiya | aadhaar_number | box_clean      | 79.2s   | "3597 8269 8331"                                                                                                                                              | ✓                                                                |
| Aditiya | aadhaar_number | box_distorted  | 83.6s   | "The image displays a document from the Government of India, specifically an ID card for a male individual named Aditya Sahu..." (generic caption, no number) | ✗                                                                |

|         |                |                |       |                                                                                                                                            |                                                |
|---------|----------------|----------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Aditiya | aadhaar_number | crop_clean     | 57.2s | "359782698331"                                                                                                                             | ✓                                              |
| Aditiya | aadhaar_number | crop_distorted | 61.5s | "3597 8269 8331"                                                                                                                           | ✓                                              |
| Sapna   | address        | box_clean      | 51.8s | "Sapna Bajaj, Date of Birth/D0: 24/06/1975 Female/ FEMALE"                                                                                 | ✗ box ignored                                  |
| Sapna   | address        | box_distorted  | 60.0s | "Sapa BAi Date of BANDAB: 24/06/1975 Female Female"                                                                                        | ✗ box ignored, garbled                         |
| Sapna   | address        | crop_clean     | 61.4s | "Address: W/O: Ajit Kumar Bajaj, Ward No 01, Seujpur, Gabhrupathar, OPP Tinupati Balaji Mandir, Dibrugarh West, Dibrugarh, Assam - 786001" | ✓ (minor "Tinupati"/"Tirupati" spelling slip)  |
| Sapna   | address        | crop_distorted | 61.3s | "Address: WI: Ajit Kumar Bajaji, Ward No 01, Seujur, Gathrupathar, OPP Tinupati Balaji Mandir, Dibrugarh West, Dibrugah, ASSAM - 786001"   | ✓ (minor spelling noise, full content correct) |

## 8. Part 6 — Full Results: Flash

| Card    | Field          | Variant        | Latency | Model output                                       | Verdict                                      |
|---------|----------------|----------------|---------|----------------------------------------------------|----------------------------------------------|
| Meghna  | aadhaar_number | box_clean      | 56.3s   | "Meghna Kumari, Meghna Kumari, 23/10/1997, female" | ✗ box ignored                                |
| Meghna  | aadhaar_number | box_distorted  | 56.1s   | "6606 2751 1012"                                   | ✓                                            |
| Meghna  | aadhaar_number | crop_clean     | 56.6s   | "6606 2751 1012 VID : 9131 4118 1801 1313"         | ✓                                            |
| Meghna  | aadhaar_number | crop_distorted | 57.2s   | "6606 2751 1012 VID : 9131 4118 1801 1313"         | ✓                                            |
| Rameez  | date_of_birth  | box_clean      | 56.0s   | "6124 8729 8241"                                   | ✗ box ignored, gave Aadhaar number           |
| Rameez  | date_of_birth  | box_distorted  | 58.8s   | "612487298241" repeated ~7 times (degenerate loop) | ✗ box ignored, degenerate output             |
| Rameez  | date_of_birth  | crop_clean     | 31.4s   | "Jambar teerha / DOB: 12/07/2002"                  | ✓ (DOB correct, garbled prefix)              |
| Rameez  | date_of_birth  | crop_distorted | 54.9s   | "Join us for this important event"                 | ✗ complete hallucination, unrelated to image |
| Prakash | gender         | box_clean      | 82.8s   | "8538 7535 4900"                                   | ✗ box ignored, gave Aadhaar number           |
| Prakash | gender         | box_distorted  | 85.7s   | "8538 7535 4900"                                   | ✗ same failure                               |
| Prakash | gender         | crop_clean     | 58.2s   | "प्रा / Male"                                      | ✓ (garbled Devanagari, English correct)      |
| Prakash | gender         | crop_distorted | 84.5s   | "ennis / Male"                                     | ✓                                            |
| Jasjot  | name           | box_clean      | 80.5s   | "Jasjot Singh"                                     | ✓                                            |
| Jasjot  | name           | box_distorted  | 80.7s   | "Jasjot Singh"                                     | ✓                                            |
| Jasjot  | name           | crop_clean     | 55.4s   | "Jasjot Singh"                                     | ✓                                            |
| Jasjot  | name           | crop_distorted | 55.8s   | "Jasjot Singh"                                     | ✓                                            |
| Aditiya | aadhaar_number | box_clean      | 1.3s*   | "3597 8269 8331"                                   | ✓                                            |
| Aditiya | aadhaar_number | box_distorted  | 67.8s   | "3597 8269 8331"                                   | ✓                                            |

|         |                |                |       |                                                                                                                                            |                                      |
|---------|----------------|----------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Aditiya | aadhaar_number | crop_clean     | 52.4s | "3597 8269 8331"                                                                                                                           | ✓                                    |
| Aditiya | aadhaar_number | crop_distorted | 55.3s | "3597 8269 8331"                                                                                                                           | ✓                                    |
| Sapna   | address        | box_clean      | 56.1s | "6227 0834 1277"                                                                                                                           | ✗ box ignored, unrelated number      |
| Sapna   | address        | box_distorted  | 54.7s | "Sanya Bai"                                                                                                                                | ✗ box ignored, garbled name fragment |
| Sapna   | address        | crop_clean     | 57.7s | "Address: W/O: Ajit Kumar Bajaj, Ward No 01, Seujpur, Gabhrupathar, OPP Tinupati Balaji Mandir, Dibrugarh West, Dibrugarh, Assam - 786001" | ✓                                    |
| Sapna   | address        | crop_distorted | 58.9s | "Address: WI0: Ajit Kumar Bajai, Ward No 01, Seuphir, Gathnupathar, OPP Tinupati Balaji, Mandir, Dibrugah West, Dibrugah, Assam - 786001"  | ✓                                    |

\* This run hit the server's prompt cache (identical image+prompt from an earlier request before a mid-benchmark USB disconnect); it is a cache-hit artifact, not a genuine cold-run latency figure. Excluded from the latency averages in Part 8.

## 9. Part 7 — Accuracy Analysis

|          | Box case (12 runs) | Crop case (12 runs) | Overall (24 runs) |
|----------|--------------------|---------------------|-------------------|
| Nano     | 4/12 (33.3%)       | 12/12 (100%)        | 16/24 (66.7%)     |
| Flash    | 5/12 (41.7%)       | 11/12 (91.7%)       | 16/24 (66.7%)     |
| Combined | 9/24 (37.5%)       | 23/24 (95.8%)       | 32/48 (66.7%)     |

### 9.1 The dominant box-case failure mode

Of the 15 combined box-case failures, 13 follow the identical pattern: **the model ignores the box and outputs a different field's value instead** — almost always name, DOB, gender, or the Aadhaar number, regardless of which field the box actually marks. This happened on both models, on both clean and distorted images, and persisted after every box-measurement correction in Part 4. The remaining 2 failures (Aditiya box\_distorted on Nano, Rameez crop\_distorted on Flash) were different failure modes — a generic image caption and a complete unrelated hallucination, respectively — but both still represent the model failing to follow the specific extraction instruction given.

### 9.2 The one crop-case failure

Flash's `rameez_crop_distorted` produced "Join us for this important event" — a complete hallucination bearing no relation to the image content. This is the only crop-case failure across 24 runs on this model, and the only one across 24 on Nano. Rameez's card is also the one outlier in the dataset (a full A4 enrollment letter, 860×1912px, a very different aspect ratio and content density from the other five compact cards) — worth further testing with more examples of this document format before concluding whether this is a Flash-specific weakness or a document-format-specific one.

### 9.3 The secondary Devanagari-transliteration pattern

Multiple crop-case responses (Rameez, Prakash, both models) show a consistent secondary quirk: the model's transliteration/rendering of the accompanying Devanagari script is frequently garbled (e.g. "पुरsey", "hells", "प्रश्न", "ennis" in place of "पुरुष") even when the requested English value is correct. This is a known, separately-documented behavior of this model family unrelated to the box/crop investigation, and did not affect scoring since the requested field's actual value was always intact.

## 10. Part 8 — Latency Analysis

|       | Box case avg                                       | Crop case avg            | Overall avg |
|-------|----------------------------------------------------|--------------------------|-------------|
| Nano  | 70.0s (range 51.8-91.3s)                           | 60.7s (range 32.1-87.0s) | 65.3s       |
| Flash | 66.9s* (range 54.7-85.7s, excl. cache-hit outlier) | 56.5s (range 31.4-84.5s) | 61.5s*      |

\* Excludes the one cache-hit artifact noted in Part 6.

### 10.1 Crop is consistently faster than box

On both models, the crop case averaged meaningfully faster than the box case (Nano: 60.7s vs 70.0s, ~13% faster; Flash: 56.5s vs 66.9s, ~15% faster). This is consistent with the root-cause finding in Part 1: vision-encoder cost scales with the tiling grid selected for the image's dimensions/aspect ratio, and the much smaller cropped images require fewer tiles. Rameez's crop-case runs were the fastest in the entire dataset (31-56s) precisely because his crop is a small, simple strip cut from an otherwise enormous source image.

### 10.2 Flash is not a meaningful latency win over Nano

Flash's overall average (61.5s) is only about 6% faster than Nano's (65.3s) — a small, likely within-noise difference, not the dramatic improvement its name might suggest. Given Flash's larger combined-accuracy profile is identical to Nano's (both 66.7% overall) and its per-case profile trades a slightly better box-case rate for a slightly worse crop-case rate, there is no clear latency-driven reason to prefer Flash over the stock model for this architecture.



## 11. Part 9 — Prompt Ablation: Can Better Wording Fix the Box Case?

Given the dominant box-case failure pattern, the natural question is whether a more carefully worded prompt could fix it. Two alternative prompts were tested (Nano only) against the three worst box\_clean failures found up to that point in the investigation.

**Forceful:** "There is a red rectangle drawn on this image, marking one specific area. Read ONLY the text inside that red rectangle. You must NOT output a person's name, date of birth, gender, address, or ID number unless that exact text is the content physically inside the red rectangle. If you find yourself about to output any field that is not inside the box, stop and look again. Output only the text found inside the box, nothing else."

**Short:** "Read only what is inside the red box. Output just that text, nothing else."

| Card    | Prompt   | Output                                                        | Verdict                                                             |
|---------|----------|---------------------------------------------------------------|---------------------------------------------------------------------|
| Meghna  | Forceful | "Meghna Kumar reddy, 23/10/1997"                              | ✗ identical failure mode                                            |
| Meghna  | Short    | "Meghna Kumar reddy, 23/10/1997, female"                      | ✗ same, worse (gender also leaked)                                  |
| Prakash | Forceful | "The answer is 8538 7535 4900."                               | ✗ identical failure mode                                            |
| Prakash | Short    | Rambling generic caption about "a passport-style document..." | ✗ different, arguably worse failure — abandoned extraction entirely |
| Sapna   | Forceful | "Sapna Bajaj"                                                 | ✗ box ignored, just the name                                        |
| Sapna   | Short    | Full correct address (minor spelling drift)                   | ✓ the one success                                                   |

**Result: 1 of 6 (17%)** — no better than the baseline box-case rate. The explicit, forceful prompt that directly named and forbade the exact failure mode observed did *no better* than a plain one-line instruction — it failed identically on 2 of 3 cases and produced a different, arguably worse failure on the third (abandoning extraction for a generic image caption). The one success came from the *short* prompt, which does not support "more explicit wording helps" as a working theory.

**Conclusion: this is not a promptable problem.** The box-ignoring behavior is a hard, trained bias in the model's fine-tuning toward its default field-extraction shape, not a matter of instruction clarity. No further prompt-engineering effort should be spent trying to fix the box case; resources should go toward the crop-based approach instead.

## 12. Part 10 — Nano vs. Flash: Head-to-Head

| Metric               | Nano                                          | Flash                                                    | Winner                                                                                            |
|----------------------|-----------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Overall accuracy     | 66.7%                                         | 66.7%                                                    | Tie                                                                                               |
| Box-case accuracy    | 33.3%                                         | 41.7%                                                    | Flash (marginally)                                                                                |
| Crop-case accuracy   | 100%                                          | 91.7%                                                    | Nano                                                                                              |
| Overall avg latency  | 65.3s                                         | 61.5s                                                    | Flash (marginally, ~6%)                                                                           |
| Worst single failure | Generic image caption (Aditiya box_distorted) | Complete unrelated hallucination (Rameez crop_distorted) | Nano (its worst failure is less severe — a caption is at least document-relevant; Flash's is not) |

Neither model is a clear winner. Flash follows box instructions somewhat more often but at the cost of a less reliable crop case (including the one true hallucination in the entire 48-run dataset) and offers no meaningful latency advantage. **Given the crop-based architecture is the one being recommended regardless (Part 11), and Nano is perfect on crop-case accuracy across all six cards, there is no evidence here to justify migrating from stock Nano to Flash.**

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## 13. Part 11 — Final Recommendations

1. **Do not send a box-annotated full image to the model.** This was the core architecture proposal under test, and it fails 60%+ of the time regardless of model, image quality, or prompt wording. The model's default field-extraction training bias overrides the visual instruction too often to be production-usable.
2. **Do send a physically cropped region instead.** The OCR layer should crop the source image to the detected field's bounding box plus a small context margin (this benchmark used 14px, which worked well even on failed OCR detections that were subsequently corrected — see Part 4), and send only that crop with a plain "extract the text in this image" instruction. This was correct on 23 of 24 runs across both models, and is also faster (fewer vision-encoder tiles on a smaller image).
3. **Do not rely on the model interpreting coordinates in text form, in either direction.** Part 2 established no usable grounding capability — coordinates given to the model as text are not reliably actionable, and asking the model to produce coordinates is even less reliable.
4. **Stay on stock Nano; there's no evidence to switch to Flash** for this use case. Flash's marginal box-case improvement doesn't matter once box-based extraction is dropped from the architecture, and its crop-case reliability and latency are not meaningfully better.
5. **Open item — Rameez's document format:** the one true hallucination in the entire dataset occurred on the one card with a substantially different format/aspect ratio (full enrollment letter vs. compact card). This should be tested with more examples of that format before concluding whether it's a real weak spot or a single outlier.
6. **Open item — batch size tuning:** raising `-ub` (ubatch size) above the default 512 (our prompts regularly exceed this) was flagged in the original latency investigation as an untested lever and remains untested here.

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## Appendix A — Exact Prompts & Server Configuration

See the companion file `prompts_used.txt` in the results folder for the complete, verbatim text of every prompt used across this investigation (main benchmark, ablation, and the earlier grounding tests), plus the exact server configuration and model files used.

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## Appendix B — Field Assignments & Ground Truth

See Part 3, §5.1 above for the full table. Raw ground-truth values and both the original and corrected box/crop pixel coordinates for all six cards are recorded in `manifest.json` in the results folder.

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## Appendix C — Raw File Index

All 24 generated test images (per model run — the same 24 images were used for both Nano and Flash), both models' full result JSON files, the prompt ablation results, and this report are located at:

```
/home/kaarunya/Downloads/model_testing/aadhaar_bench_results/

manifest.json           - ground truth + box coordinates (post-correction) for all 6
cards
nano_results.json       - all 24 Nano runs, full raw model output + latency
flash_results.json      - all 24 Flash runs, full raw model output + latency
prompt_ablation_results.json - all 6 ablation runs
prompts_used.txt        - every prompt used, verbatim
{name}_box_clean.jpg    - per card, 6 files
{name}_box_distorted.jpg - per card, 6 files
{name}_crop_clean.jpg   - per card, 6 files
{name}_crop_distorted.jpg - per card, 6 files
aadhaar_benchmark_report.pdf - this report
```

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VisionPsy-Nano / Flash Box-vs-Crop Benchmark Report — generated 2026-09-18. Contains real personal data (Aadhaar numbers, addresses, names, photographs) belonging to six individuals whose ID card images were supplied for this QA testing workflow. This document is stored locally only and has not been published or uploaded to any external or shared service.